

ml. of IR-120 resin. The columns were washed with 150 ml. of 2*M* hydrochloric acid and spot tests showed that the ions were being eluted. The thorium was then eluted, and analysis of the eluate showed no error, within experimental limits, from the presence of the respective interferences.

The evaluation of interferences was limited to the examination of elements which might be in the samples to be analyzed. The cation exchange separation procedure should provide some separation from trivalent yttrium, lanthanum, and rare earths. However, the quadrivalent elements such as zirconium and hafnium would tend to follow thorium closely.

Results. As there was no standard uranium ore of known thorium concentration available, a known amount of standard thorium solution was added to aliquots of solutions from four different types of uranium ores. Aliquots of the ore solutions, with and without added thorium, were analyzed by the recommended procedure. The results of this test, shown

Table III. Test of Cation Exchange-Spectrophotometric Method

Ore Type	Thorium, Mg.		Recovered
	Added	Found	
Lignite	0.489	0.514	0.489
	None	0.025	
Dyson Rum	0.489	0.488	0.488
Jungle	None	None	
Lukachukai	0.489	0.474	0.474
	None	None	
Uranium concentrate	0.489	0.479	0.479
	None	None	

in Table III, indicate a good recovery of thorium.

LITERATURE CITED

- (1) Bane, R. W., U. S. Atomic Energy Comm., **CC-3336** (1945).
- (2) Banks, C. V., Byrd, C. H., *ANAL. CHEM.* **25**, 416 (1953).
- (3) Brown, W. E., Rieman, W., *J. Am. Chem. Soc.* **74**, 1278 (1952).
- (4) Cuttitta, F., U. S. Geol. Survey **TEI-498** (1955).

- (5) Dyrssen, D., *Svensk Kem. Tidskr.* **62**, 153 (1950).
- (6) Gordon, L., Firsching, F. H., Shaver, K. J., *ANAL. CHEM.* **28**, 1476 (1956).
- (7) Guest, R. J., *Can. Dept. Mines and Tech. Surveys, Topical Rept. TR-98* (1952).
- (8) Kraus, K. A., Moore, G. E., Nelson, F., *J. Am. Chem. Soc.* **78**, 2692 (1956).
- (9) Kraus, K. A., Nelson, F., *Proc. Intern. Conf. Peaceful Uses Atomic Energy, Geneva 1955* **7**, 113 (1956).
- (10) Menis, O., Manning, D. L., Goldstein, G., *ANAL. CHEM.* **29**, 1426 (1957).
- (11) Radhakrishna, B. P., *Anal. Chim. Acta* **6**, 351 (1952).
- (12) Samuelson, O., "Ion Exchangers in Analytical Chemistry," p. 35, Wiley, New York, 1953.
- (13) Smith, M. E., U. S. Atomic Energy Comm., **LA-1897** (1955).
- (14) Thomason, P. F., Perry, M. A., Byerly, W. M., *ANAL. CHEM.* **21**, 1239 (1949).

RECEIVED for review July 11, 1957. Accepted February 16, 1958. The Raw Materials Development Laboratory is operated by the National Lead Co., Inc., for the U. S. Atomic Energy Commission. Work carried out under Contract No. AT(49-6)-924.

Chromatography of Amino Acids on Sulfonated Polystyrene Resins

An Improved System

STANFORD MOORE, DARREL H. SPACKMAN, and WILLIAM H. STEIN

The Rockefeller Institute for Medical Research, New York 21, N. Y.

►Improved procedures have been developed for the chromatographic determination of amino acids on columns of finely pulverized 8 % cross-linked sulfonated polystyrene resins. The use of a smaller particle size has permitted faster flow rates, and appropriate choice of eluents has simplified operations. Complete analyses of protein hydrolyzates can be performed in about 48 hours with fraction collectors and in 24 hours with automatic recording equipment. The same system, with minor modifications, can be used for determination of amino acids and related compounds in blood plasma, urine, and animal tissues.

MODIFICATIONS in the earlier ion exchange procedures (8, 9) make possible a complete amino acid analysis of a peptide or protein hydrolyzate in 24 to 48 hours. The modified system can be used either with a fraction col-

lector or with automatic recording equipment (12, 13). The method (in its manual and automatic forms) has been used for analyses reported in recent studies on histones (1), hemoglobin (16), and ribonuclease (5).

In order to use a faster flow rate without undue broadening of the peaks (6), a resin of smaller particle size has been utilized. Improved resolving power and speed have been attainable with columns packed with a very finely pulverized 8% cross-linked sulfonated polystyrene resin; Amberlite IR-120 in "micropowder" form has shown good reproducibility from batch to batch over a 3-year period of test. The commercial powder has been classified into fractions possessing the desired ranges of particle size by the hydraulic method recently described by Hamilton (2).

The mode of operation of the columns combines features drawn from both the Dowex 50-X8 (8) and 50-X4 (9)

methods. The neutral and acidic amino acids are separated on a 150-cm. column of Amberlite IR-120, and the basic amino acids are applied to a 15-cm. column. The two columns may be used repeatedly without having to be repoured. A single temperature of operation (50° C.) has proved possible. Only two buffers are required with the 150-cm. column, necessitating one change of eluent. A single buffer is employed for the elution of the basic amino acids from the 15-cm. column, making possible a stable base line throughout the analysis. The effluent curves obtained when a synthetic mixture of amino acids is analyzed by this procedure are shown in Figure 1.

Except where noted, the directions in this manuscript apply when the columns are used with fraction collectors.

EXPERIMENTAL

Resin. The finely ground sodium

salt of Amberlite IR-120 (Rohm & Haas Co.) is marketed through The Fisher Scientific Co., 635 Greenwich St., New York 14, N. Y., in three grades suitable for chromatographic use bearing the following dry-sieved specifications: Amberlite IR-120 (CG-120), Type I (100-200 mesh), Type II (finer than 200 mesh; Catalog No. A-530), and Type III (400-600 mesh; Catalog No. A-932). [A similar Type III grade of the carboxylic acid resin Amberlite IRC-50 (CG-50) from the same source (Catalog No. A-933) gives high yields of through-200-mesh particles suitable for the chromatography of peptides and proteins (cf. 7).]

The Type II Amberlite IR-120 has usually given the highest yields of material of appropriate size for the preparation of columns to be employed with fraction collectors; the Type III product is required for the recording method, where columns packed with smaller particles are needed to permit the use of the faster flow rates. One pound of the suitable type of resin usually supplies enough material for two 0.9×150 cm. columns and one 0.9×15 cm. column, but it is desirable to have on hand at least 2 pounds of starting material.

Separation of Particles of Appropriate Size. One pound of the dry sodium salt of the resin, as purchased, is transferred in a hood to a jar containing about 10 liters of water. Resin present in any foam that remains after the initial stirring of the mixture can be transferred to a beaker, treated with a little acetone, and added to the main suspension. After thorough distribution of the resin in the water, the mixture is allowed to settle for about 6 hours. The supernatant suspension, which contains very fine particles of resin, is withdrawn by suction. The resin is suspended and re-settled three or four times. The product is then washed slowly on a large Büchner funnel with 2 liters of 4*N* hydrochloric acid, followed by 500 ml. of water. The moist resin is suspended in 2 liters of 2*N* sodium hydroxide, and the mixture is heated on the steam bath for 1 hour. The resin is finally collected on a Büchner funnel and washed with water until neutral.

Hydraulic separation by the method of Hamilton (2) has been carried out with apparatus which can be used with tap water (Figure 2). In general, reproducible results in the fractionation depend upon the delivery of up to 600 ml. per minute of bubble-free water at relatively constant temperature and at constant flow rate over periods of hours. Hamilton used distilled water which, when available in sufficient quantity, is preferable to tap water for two reasons. First, the presence of calcium, magnesium, and other cations is undesirable, although with New York City tap water the uptake of

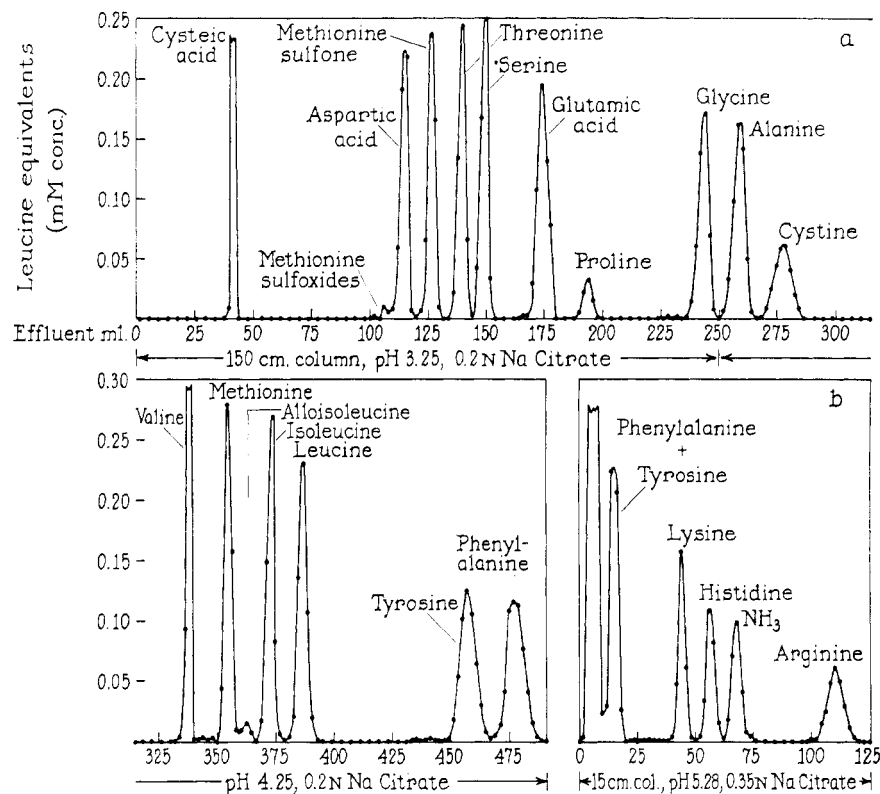


Figure 1. Chromatographic fractionation of synthetic mixture of amino acids on columns of Amberlite IR-120

Load on 0.9×150 cm. column was 1 μ mole of each amino acid (0.5 μ mole of cystine)

1a. Obtained by elution of neutral and acidic amino acids at 50° from column at flow rate of 12 ml. per hour

1b. Elution of the basic amino acids (at half the above load) from a 0.9×15 cm. column of the resin at 50° and at a flow rate of 25 ml. per hour

Effluent collected in 2-ml. fractions. Additional placements and alternative conditions for elution are referred to in Figures 2 and 9 of (12)

small amounts of such cations by the sodium salt of the resin does not change the density of the particles sufficiently to interfere with the fractionation. Secondly, when tap water is used, there is more difficulty in obtaining a stream free from bubbles of air. The deaerating system indicated in Figure 2 may not be adequate for all water lines; in some instances a larger glass wool filter may be necessary. To improve the deaeration, Kenneth Woods, Cornell Medical Center, has added two spiral condensers to the apparatus, one in the stream (cold tap water, in this instance) flowing from the reservoir into the filter and one in the down-flowing line just before the needle valve. Hot tap water is circulated through the first jacket, and water at 25° through the second.

The specific particles obtained at various flow rates will depend upon the shape and maximum diameter of the 2-liter separatory funnel (Corning No. 6400, 15.2 cm. maximum in outside diameter). With the equipment shown in Figure 2, water at $25^\circ \pm 2^\circ$ C. (blended from hot and cold tap water) is run into the glass reservoir, which serves as a constant-head device.

The water is filtered and freed of air by passage through the loosely packed section of borosilicate glass wool, which is held in place by a layer of stainless steel screen. Liberated air bubbles escape to the atmosphere through the upright vent tube. The rate at which water flows to the separatory funnel is controlled accurately by a Teflon and glass needle valve (Emil Greiner Co., 20-26 North Moore St., New York 13, N. Y., No. 10428 with No. 10428A glass connection tubes) and is measured by a flowmeter (2).

For the fractionation, 400 ml. or less of the settled resin prepared as described above is suspended in water to give a total volume of about 1 liter and transferred to the 2-liter separatory funnel. With the needle valve fully open and the tube to the bottom of the separatory funnel disconnected and directed to the drain, the temperature and the adequacy of the volume of flow of water are checked. The tube (temporarily clamped off) is then attached to the bottom of the separatory funnel, and flow at full speed (about 600 ml. per minute) is started through the funnel. When the funnel is about $3/4$ full, the rate of flow is reduced to 110 ml. per minute by means of the needle valve, and is maintained at this rate until no more particles can be seen leaving the outlet tube. The length of time before

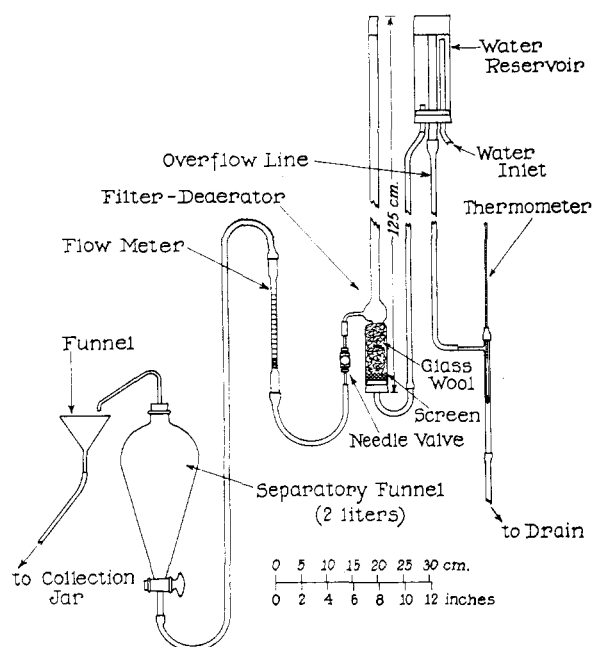


Figure 2. Equipment used for separation of resin particles by hydraulic method of Hamilton (2)

Flexible tubing is of Tygon. See text for alternative arrangements for removal of air bubbles from influent stream of water

the top part of the funnel becomes clear will vary with the sample of resin, but usually is several hours. The resin removed at this flow rate is collected in large jars and is permitted to settle out.

The flow rate is then increased to 280 ml. per minute. The resin that comes over under these conditions (fraction C, in Table I) contains particles with an average diameter of 40 microns. Similarly, fraction D is collected after the flow rate is raised to 580 ml. per minute. The material that came over at 110 ml. per minute is subfractionated, at a flow rate of 50 ml. per minute. Each fraction is finally recycled once before being used. For example, the fraction obtained initially at 110 to 280 ml. per minute (fraction C) is placed in the funnel; any resin coming over at 110 ml. is added to fraction B and any material remaining at 280 ml. is added to fraction D.

In Table I are given the results of the fractionation of lots of Amberlite IR-120, Types II and III. For comparison, the screening data for the wet particles of the two types were: Type II, through 200 mesh 92%, through 400 mesh 28%; Type III (Lot 6328) through 200 mesh 100%, through 400 mesh 75%. The resin used with the automatic recorder is composed almost entirely of particles smaller than 400 mesh, and is secured conveniently by the hydraulic method in a form free from material that is very much finer.

The final test of the suitability of a given fraction of resin is the performance of a column prepared from it. If a column is too tight, it may be

necessary to repeat the hydraulic fractionation in order to remove a small percentage of the finest particles from the resin. If the resolving power of the column is deficient, some of the coarser particles must be removed.

Fractions of resin that are to be stored for later use may be suspended in 0.2N sodium hydroxide and kept in polyethylene bottles at 4°. If moist, neutral suspensions are stored at room temperature, contamination by molds may occur.

Preparation of Ion Exchange Columns. The thick-walled chromatograph tubes required are similar in specifications to those previously employed with fraction collectors (8, 9), and are available from the Scientific

Glass Apparatus Co., Bloomfield, N. J. [Catalog No. J-1665-1B, 0.9 cm. in inside diameter with heights of 166 and 30 cm., respectively, above the sintered plates, and with water jackets to fit. Similar tubes with ground joints attached are used with the automatic recording equipment (12).]

The 150-cm. column to be used with fraction collectors is poured with resin from the fraction D obtained by hydraulic fractionation of the powdered Amberlite IR-120. The average diameter of the particles in this fraction was 56 microns (Table I). About 100 ml. of settled resin are required for each 150-cm. column.

The resin to be used is washed on a Büchner funnel with 1 liter of 4N hydrochloric acid per 200 to 300 ml. of settled resin (to remove metal ions that may have been adsorbed from the tap water during the fractionation), followed by water, and 2N sodium hydroxide until the filtrate is strongly alkaline. The sodium salt of the resin is washed with water and then with several hundred milliliters of 0.2N buffer at pH 4.25 (Table II; without BRIJ 35 or thiodiglycol) until the filtrate is about pH 4. The columns are poured with a suspension of resin in this buffer. The composition of the slurry should be such that the volume of the supernatant buffer is twice that of the settled resin. If the handling of the resin subsequent to the hydraulic fractionation has caused the formation of small amounts of very finely divided material, the resin should be settled two or three times from about 6 to 8 volumes of the buffer until the supernatant liquid is practically free of suspended particles.

The preparation of uniformly packed columns requires close adherence to the prescribed procedure. The method for preparing 0.9 × 150 cm. columns in five or six sections is similar to that described for Dowex 50-X8 (8), with the following exceptions. Before the first section is poured, the outlet of the tube is closed. As soon as the slurry of resin has been poured in, the outlet is opened and about 2 cm. of resin bed is

Table I. Description of Fractions of Amberlite IR-120 Prepared by Hydraulic Fractionation by Procedure of Hamilton (2)

Fraction	Water Flow (2-Liter Funnel), Ml./Min.	Use of Resin	Particle Diameter ^a , μ	Yield, %		
				Type II	Lot 6328	Lot 123
A	50	Not used	<25	7	50	6
B	50-110	Automatic recorder, 15- and 50-cm. columns	25-30			13
C	110-280	Automatic recorder, 150-cm. column	40 ± 7	21	32	37
D	280-580	Fraction collector, 15- and 50-cm. columns				
E	Residue	Fraction collector, 150-cm. column	56 ± 9	40	18	44
		Not used	>70	32	0	0

^a Measured by method of Martin, Blyth, and Tongue [cf. Hamilton (2)].

^b From resin freed of very finely divided material removable by four settlings from water.

Table II. Preparation of Sodium Citrate Buffers

pH	Sodium Concn., <i>N</i>	Citric Acid H ₂ O, G.	NaOH (97%), G.	HCl (Concd.), Ml.	Final Vol., Liters	Phenol Added to Final Vol., G.	Fraction Collector		Automatic Recorder	
							TG ^a	BRIJ ^b	TG ^a	BRIJ ^b
3.25 ± 0.01 ^c	0.20	840	330	426	40	40	10	5	5	2
4.25 ± 0.02 ^c	0.20	840	330	188	40	40	10	5	5	2
5.28 ± 0.02 ^c	0.35	491	288	136	20	20	..	5	..	2
4.26 ± 0.02 ^c	0.38	532	312	307	20	20	..	5	..	2
6.50 ± 0.05	0.38	133	78	6.3	5	5	..	5	..	2

^a Thiodiglycol (TG), purified [(9) Table I], available from Pierce Chemical Co., P. O. Box 117, Rockford, Ill.

^b BRIJ 35 detergent solution, prepared as described in (9), footnote 4. Both TG and BRIJ 35 solution are added in stated quantities to several liters of buffer shortly before use, not to 40-liter quantities to be stored. BRIJ 35 can be omitted if it interferes with action of fraction-collecting device.

^c If adjustment of pH is necessary, addition of 1 ml. of 50% NaOH or 2 ml. of concd. HCl causes a change of about 0.01 pH unit.

allowed to form under gravity flow. Air pressure of 30 cm. of mercury is then applied at the top of the tube. The pouring is done at room temperature. After the resin has settled to constant height, all of the supernatant buffer is withdrawn by suction and the next section is poured on top of the firm surface. After the third section has been added, a 0.9 × 40 cm. extension tube is attached to the top of the chromatograph tube by means of a butt joint held by rubber tubing about 0.8 cm. in inside diameter (or through 18/9 semiball joints). The final height of the resin should be several centimeters greater than 150 cm., to allow for some further packing during operation.

Before use, a new 150-cm. column should be washed with 0.2*N* sodium hydroxide and equilibrated with boiled buffer at pH 3.25, as described below for the regeneration of the columns. Both operations are carried out at room temperature before the temperature is raised to 50° (Aminco No. 4-96 bath). The rate of flow at 50° should then be checked. If the column is too tight to permit delivery of 12 ml. of effluent per hour under 20- to 40-cm. pressure, the resin is withdrawn [(9), p. 899] from the tube and some of the finest particles must be removed from the resin by settling or by refractionation, and the column repoured. If the column permits a much faster flow rate than 12 ml. per hour at 20-cm. pressure, a test chromatogram may be run, but the column is likely to be too loose to give the desired resolving power. Hydraulic separation to remove more of the larger particles is then required. Once a column of resin possessing the desired particle size is obtained, it can be used indefinitely; adjustment of particle size distribution is thus not a recurring task.

The 15-cm. columns are poured similarly, in one or two sections, with resin that has been slurried in 0.35*N* buffer at pH 5.28 (Table II). All columns should be prepared in buffer with the same ionic strength as that with which they will be operated; if 50-cm. columns are poured for analysis of physiological fluids, the buffer should be 0.38*N*, pH 4.26 (Table II). As the resistance to flow through the 15-cm. columns is never great, it is advantageous to pour these from a small amount of fraction C resin (Table

I, average diameter 40 microns), to obtain high resolving power. The column is equilibrated before use with boiled buffer that contains BRIJ 35.

Buffers. The first three buffers listed in Table II are used in the operation of the two columns. A small amount of buffer at pH 2.2 (9) is required for adjustment of the pH of the samples to be analyzed. The first two buffers are conveniently made up in 40-liter quantities. The solutions were stirred in a 15-gallon polyethylene tank (with spigot), covered, and allowed to stand overnight before the pH was checked and adjusted, if necessary. The solutions have been stored at 4° in 13-gallon polyethylene carboys equipped with needle-type valves near the base. The transfer from the tank at bench level to the carboy at floor level was made through a tube containing a sintered-glass filter (J-5010, extra coarse, 60 cm., Scientific Glass Apparatus Co., Inc.). The other buffers were stored in 5-gallon polyethylene carboys. To prevent growth of mold during storage of the buffers, 0.1% phenol was added (14).

Buffer for immediate use is kept in a 500-ml. separatory funnel fitted with a stopcock clamp and connected to the top of the column (9). To prevent formation of bubbles in the resin bed, which would occur as a result of the release of air when the buffers are heated to 50°, 500 ml. of the buffer are brought to a boil and added while hot to a separatory funnel. About 50 ml. of mineral oil are layered over the buffer. A long-stemmed funnel is employed to introduce subsequent samples of boiled buffer below the level of the oil.

Operation of 150-Cm. Columns.

The procedure for addition of the sample to the column at 50° is similar to that for columns of Dowex 50-X8 (8) and 50-X4 (9).

The optimum load is 0.5 to 1.0 μ mole per amino acid, or 1 to 2 mg. of hydrolyzed protein. The initial buffer is at pH 3.25, the column is maintained at 50° throughout the analysis, and the effluent is collected in 2-ml. fractions, which are analyzed photometrically by addition of 1 ml. of the modified ninhydrin reagent (10). Ninhydrin (reagent grade) and hydroindantin (anhy-

drous) are currently purchased from the Pierce Chemical Co., P. O. Box 117, Rockford, Ill. Methyl Cellosolve, peroxide-free grade, is obtained directly from the manufacturer (Union Carbide Chemicals Co.). The filtered 4*N* sodium acetate buffer, made in 10-liter quantities, can be stored at room temperature without a preservative. The stability of the ninhydrin reagent solution on storage can be increased by wrapping the reservoir completely with black tape, to exclude light. A narrow vertical slit is cut in the covering to permit the level in the reservoir to be observed.

An eluent at pH 4.25 is introduced at a time (cf. Figure 1, *a*) designed to allow valine to emerge with the new buffer. The change is made at an effluent volume 2.15 times that at which the aspartic acid peak has emerged, or about 260 ml. At a flow rate of 12 to 14 ml. per hour, this point will have been reached on the day after the column has been started. By the following morning, tyrosine and phenylalanine will have emerged. The amino acids may all be eluted somewhat more slowly or more rapidly than indicated in Figure 1, if the chromatograph tube has a diameter slightly different from 0.9 cm.

The position of cystine is extremely sensitive to pH [(8), p. 669], and the user must determine by experiment what the exact pH value of the initial buffer should be, so that cystine will be eluted about midway between alanine and valine. Subsequent lots of the buffer should be adjusted as closely as possible to the pH value, using as reference 0.05*M* potassium acid phthalate at pH 4.00.

To prepare the 150-cm. column for re-use, the basic amino acids retained by the resin are eluted with 0.2*N* sodium hydroxide (relatively free from carbon dioxide, containing 5 ml. of BRIJ 35 solution per liter). The alkali can be allowed to run through under gravity flow overnight, or under a few centimeters of pressure until the visible advancing front of sodium hydroxide is more than halfway down the column. The sodium hydroxide solution is not boiled or stored under oil; the funnel should be stoppered when not in use, so as to minimize uptake of carbon

dioxide. The column is re-equilibrated for the next analysis by passing through it overnight at room temperature or 50° about 120 ml. of buffer at pH 3.25. If two 150-cm. columns are on hand, one can always be ready for use.

Operation of 15-Cm. Column. A 15-cm. column can be operated on the fraction collector during the day, after a 150-cm. column has been removed. For determination of the basic amino acids, the aliquot taken is equal to or one half the size of the sample used for the 150-cm. column. The pressure used during the addition of the sample should not exceed 10 cm. Elution is carried out at 50° with a buffer at pH 5.28 and a rate of 25 to 30 ml. per hour (Figure 1, *b*). The analysis is complete after about seventy 2-ml. fractions have been collected, which requires only 5 to 6 hours. By the elution of lysine, histidine, ammonia, and arginine with a single buffer, it is possible to obtain a constant base line throughout the chromatogram. When a change in eluent is made (8, 9), ammonia accumulated by the resin from the preceding buffer may be eluted and give rise to a plateau before or after the ammonia peak. In the present procedure the buffer used to fill the top part of the chromatograph tube after the sample has been added should be drawn from the separatory funnel containing the buffer with which the column is being operated. About 80 ml. of this same buffer should first be run through the column if the funnel has just been refilled. Even a few milliliters of buffer that contains a concentration of ammonia higher or lower than the main lot can give rise to a shoulder or valley following the ammonia peak. The columns can normally be used repeatedly without regeneration, because most samples do not contain any ninhydrin-positive constituents with a greater retardation than arginine. Regeneration, if required, can be accomplished with 10 ml. of 0.35*N* sodium hydroxide containing BRIJ 35, after which 80 ml. of the pH 5.28 buffer should be run through before the column is re-used.

The position of histidine is fairly sensitive to pH, and the pH of the buffer may need adjusting to center the peak between those of lysine and ammonia.

RESULTS

Recoveries. As was the case for the Dowex 50-X4 method (9), analyses of known mixtures have yielded recoveries of $100 \pm 3\%$ for all the amino acids given in Figure 1 except methionine, for which the recovery is 95%. This value is higher than the 90% obtained in the earlier method, because of the higher concentration of thiodiglycol in the buffers now used. The base line

for the integration of the first peaks emerging with the buffer at pH 4.25 should be taken after the leucine peak. To obtain a stable base line, it is essential to prevent effluent fractions from taking up ammonia from the air [(19), p. 909].

Resolution. The conditions of pH and temperature chosen for the chromatogram illustrated in Figure 1 have been designed to separate in a minimum of time the constituents present in the acid hydrolyzates of most proteins. By starting the 150-cm. column of Amberlite IR-120 at 30° instead of 50°, a number of useful improvements in resolution are obtained. The same buffers are used. The results are shown in Figure 9, *a*, of (12). The temperature is raised to 50° at the same time that the buffer is changed to pH 4.25; this change is made at 2.40 times the position of the aspartic acid peak when the columns are being operated on fraction collectors. At a rate of 16 ml. per hour, the chromatogram can be completed in 2 days.

When the 150-cm. columns are employed in this manner with fraction collectors to determine the neutral and acidic amino acids in blood plasma, the sequence of the peaks through tyrosine and phenylalanine is almost identical to that obtained with Dowex 50-X4 (17). A protein-free filtrate from 4 ml. of human plasma is an adequate sample for each column. A 1- to 2-ml. sample of human urine (15) can be analyzed in the same manner.

When tissue extracts are chromatographed (18), the presence of glutathione presents an additional problem. Glutathione is eluted near glutamic acid at 30°, and near aspartic acid at 50°. For some purposes, the presence of glutathione in a somewhat broad zone underlying the glutamic acid peak would not be a disadvantage (19), but this interference can be eliminated if the glutathione is converted to the *S*-sulfonate by reaction with sulfite. The conditions for effecting this conversion have been studied in this laboratory by J. L. Bailey.

In a typical experiment, 0.5 ml. of an extract derived from 0.4 gram of brain tissue was brought to pH 7.2 to 7.5 with sodium hydroxide and allowed to stand for 4 hours at room temperature to permit autooxidation of reduced glutathione to the *S*-*S* compound. After addition of 0.2 ml. of 0.5*M* sodium sulfite the solution was allowed to stand for 1 hour, and then brought to pH 1.5 by the dropwise addition of 1*N* hydrochloric acid. Upon chromatography, glutathione-*S*-sulfonate (along with any cysteine-*S*-sulfonate present) emerges at about 40 effluent ml., just in advance of the glycerophosphoethanolamine position. If any reduced glutathione remains in the reaction

mixture, it will give rise to a peak in front of that of aspartic acid.

Improved separation of the principal basic amino acids and related compounds in blood plasma, urine, and tissue extracts can be obtained with a column 50 cm. high. This column may be prepared from fraction C of Amberlite IR-120 (Table I); with particles of this size excellent resolution is secured at a flow rate of 16 to 20 ml. per hour maintained by an air pressure of 25 to 40 cm. of mercury. The column is operated initially at 30°, and 2-ml. fractions are collected. Elution is begun with a 0.38*N* buffer at pH 4.26 (Table II) and carried beyond the ammonia peak without change in order to ensure a constant base line. The separation obtained is shown in Figure 9, *b*, of (12). The temperature is raised to 50° just after the position of emergence of carnosine, in order to elute arginine more rapidly. A simultaneous change to a 0.38*N* buffer at pH 6.5 (Table II) will bring arginine off the column even earlier (at about 500 ml.). This additional buffer is not used with the recorder, and is not required in the fraction collector method unless it is necessary to try to complete the chromatogram in 24 to 30 hours; the analysis with the 50-cm. column is usually performed over a 2-day period on the fraction collector. If the second buffer is used, the column must be equilibrated with the pH 4.26 buffer (200 ml.) before re-use. If a rapid determination of arginine is required, the resolving power of the 15-cm. column (Figure 1) provides complete separation of this amino acid from the other ninhydrin-positive constituents of physiological fluids.

The desired spacing of the peaks from the 50-cm. column is dependent upon the exact pH of the eluent. At pH 4.42 instead of pH 4.26 lysine and ammonia emerge together. Therefore, adjustment of the pH of the buffer may be required after an initial test run.

DISCUSSION

Pressure and Flow Rate. Several considerations limit the maximum pressures which can conveniently be applied to the columns. Pressure in excess of 40 cm. on the 500-ml. separatory funnels is to be avoided, for reasons of safety. Flow rates up to 30 ml. per hour (47 ml. per hour per sq. cm. cross-sectional area of the column) can be used with a fraction collector, as they are used with automatic recording equipment (12), if suitable operating conditions are adhered to—i.e., the column is prepared from a sufficiently finely divided resin (Table I), a pump rather than air pressure is used to drive the deaerated

buffer through the column, and the chromatograph tubes are equipped with semiball connections.

Columns of Dowex 50-X4 (9) also could be operated at faster flow rates in a similar manner. The separation of peptides on columns of Dowex 50-X2 (4, 7) can be speeded up by using a pump, which should be placed in the line between the reservoir bottle or mixing chamber and the column. Chromatograph tubes suitable for use with higher pressure (12) are then needed. The 2% cross-linked resin is available in a through-400-mesh grade (Technical Service and Development, Dow Chemical Co., Midland, Mich.) which will give good yields of material with small enough particle size to permit a flow rate of 30 ml. per hour per sq. cm. cross-sectional area of the column.

Allo Forms of Hydroxylysine and Isoleucine. Hamilton and Anderson (3) and Piez (11) observed that racemization at the α -carbon of hydroxylysine occurs during acid hydrolysis and gives rise to appreciable amounts of allohydroxylysine, which appears as a separate peak on effluent curves from columns of Dowex 50-X8. Piez (11) has also shown that allosioleucine emerges ahead of isoleucine and at a position overlapping that of methionine from 100-cm. columns of Dowex 50-X8 (8). In the present method this overlap is eliminated (Figure 1), and allo-

isoleucine appears in a peak emerging between methionine and isoleucine. Any trace of allosioleucine obtained as a result of racemization at the α -carbon of isoleucine during hydrolysis can thus be measured. When the fraction collector method is used, the amount of allosioleucine in protein hydrolyzates is frequently too small to be detected. The recording method (12), however, is capable of demonstrating the trace of allosioleucine present (up to 3%) after 24 hours of hydrolysis. After 72 hours of hydrolysis, the allosioleucine formed may correspond to as much as 5% of the isoleucine, and the sum of both peaks probably should be used to calculate the isoleucine content of the protein, in the same manner that the methionine sulfoxides which may be formed during hydrolysis are included in calculating the methionine content of the protein.

ACKNOWLEDGMENT

The authors wish to acknowledge the technical assistance of Renate Mikk and Kerstin Johansson during these experiments. They are greatly indebted to P. B. Hamilton for sharing the details of his procedure during its developmental stages.

LITERATURE CITED

- (1) Crampton, C. F., Stein, W. H., Moore, S., *J. Biol. Chem.* **225**, 363 (1957).

- (2) Hamilton, P. B., *ANAL. CHEM.* **30**, 914 (1958).
- (3) Hamilton, P. B., Anderson, R. A., *J. Biol. Chem.* **213**, 249 (1955).
- (4) Hirs, C. H. W., Moore, S., Stein, W. H., *Ibid.*, **219**, 623 (1956).
- (5) Hirs, C. H. W., Stein, W. H., Moore, S., IUPAC Symposium on Protein Structure, Wiley, New York, in press.
- (6) Mayer, S. W., Tompkins, E. R., *J. Am. Chem. Soc.* **69**, 2866 (1947).
- (7) Moore, S., Stein, W. H., *Advances in Protein Chem.* **11**, 191 (1956).
- (8) Moore, S., Stein, W. H., *J. Biol. Chem.* **192**, 663 (1951).
- (9) *Ibid.*, **211**, 893 (1954).
- (10) *Ibid.*, p. 907.
- (11) Piez, K. A., *Ibid.*, **207**, 77 (1954).
- (12) Spackman, D. H., Stein, W. H., Moore, S., *ANAL. CHEM.* **30**, 1190 (1958).
- (13) Spackman, D. H., Stein, W. H., Moore, S., *Federation Proc.* **15**, 358 (1956).
- (14) Spedding, F. H., Fulmer, E. I., Butler, T. A., Gladrow, E. M., Gobush, M., Porter, P. E., Powell, J. E., Wright, J. M., *J. Am. Chem. Soc.* **69**, 2812 (1947).
- (15) Stein, W. H., *J. Biol. Chem.* **201**, 45 (1953).
- (16) Stein, W. H., Kunkel, H. G., Cole, R. D., Spackman, D. H., Moore, S., *Biochim. et Biophys. Acta* **24**, 640 (1957).
- (17) Stein, W. H., Moore, S., *J. Biol. Chem.* **211**, 915 (1954).
- (18) Tallan, H. H., Moore, S., Stein, W. H., *Ibid.*, **211**, 927 (1954).
- (19) *Ibid.*, **230**, 707 (1958).

RECEIVED for review February 28, 1958.
Accepted May 17, 1958.

Automatic Recording Apparatus for Use in the Chromatography of Amino Acids

DARREL H. SPACKMAN, WILLIAM H. STEIN, and STANFORD MOORE
The Rockefeller Institute for Medical Research, New York 21, N. Y.

► Quantitative determination of amino acids is made simpler and more rapid by an instrument for automatically recording the ninhydrin color value of the effluent from ion exchange columns. The influent buffer, freed of air, is pumped at a constant rate through a column of sulfonated polystyrene resin. The effluent is met by a capillary stream of ninhydrin reagent delivered by a second pump. The color is developed by passing the mixture of reagent and effluent through a spiral of capillary Teflon tubing immersed in a boiling water bath. The absorbance of the resulting solution is measured continuously at 570 and 440 m μ as it flows through a cylindrical glass cell of 2-mm. bore. The peaks on the recorded curves can be inte-

grated with a precision of $100 \pm 3\%$ for loads from 0.1 to 3.0 μ moles of each amino acid. A hydrolyzate of a protein or peptide may be analyzed in less than 24 hours. The more complex mixtures characteristic of blood plasma, urine, and mammalian tissues can be analyzed in 2 days. The instrument is applicable in principle to detection of ninhydrin-positive constituents in the effluent from various types of chromatograph columns.

QUANTITATIVE determination of amino acids bears a relationship to the chemistry of proteins and complex peptides similar to that which elementary analysis bears to the chemistry of

simpler organic molecules. In recent studies on the structure of ribonuclease, a partial structural formula for the enzyme has been deduced primarily from data obtained by the quantitative amino acid analysis of the peptides formed as a result of the action of trypsin, pepsin, and chymotrypsin (4). Determinations of amino acids are also required in investigations of the distribution of amino acids and their derivatives in physiological fluids (16, 17) and mammalian tissues (18) as well as in the analysis of foods (10, 14). To facilitate such studies, there is need for an accurate method of amino acid analysis that is more rapid and less laborious than the techniques employed heretofore. To fill this need, a recording apparatus has been constructed which,